Hereditary Cardiac Diseases

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Types of hereditary cardiac diseases

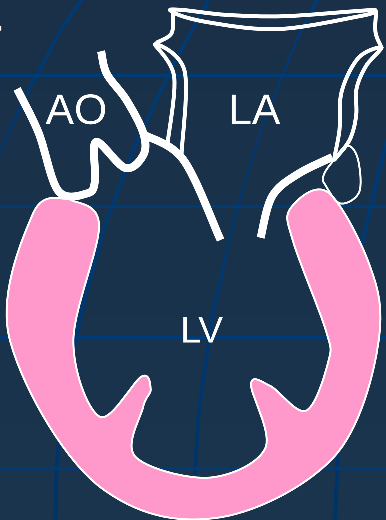
- Inherited cardiomyopathies
  - HCM, DCM, ARVD, RCM
- Inherited disorders of rhythm and conduction
  - LQTS, BRS, CPVT, PCCD
- *Familial hypercholesterolemia*
- *Connective tissue diseases*
  - *Marfan, Ehlers-Danlos, Loeys-Dietz syndrome, Familial thoracic aortic aneurysm and dissection, bicuspid valve aortopathy*
- Congenital $\neq$ Inherited
  - (VSD, ASD, WPW)

Cardiomyopathies

- A heterogeneous group of diseases of myocardium
- Exhibits inappropriate VENTRICULAR HYPERTROPHY / DILATATION / STIFFNESS
- Associated with mechanical and/or electrical dysfunction
- Confined to the heart / are a part of generalized systemic disorders
- ‘Heart Failure’ vs. ‘Cardiomyopathies’
  - HF is a clinical syndrome; CM is an anatomic and pathologic diagnosis
  - CM can be an underlying cause of a HF but CM does not always cause HF
  - HF can be caused by CM but can also be related to other cardiac structures / electrical disturbances

Morphological Classification

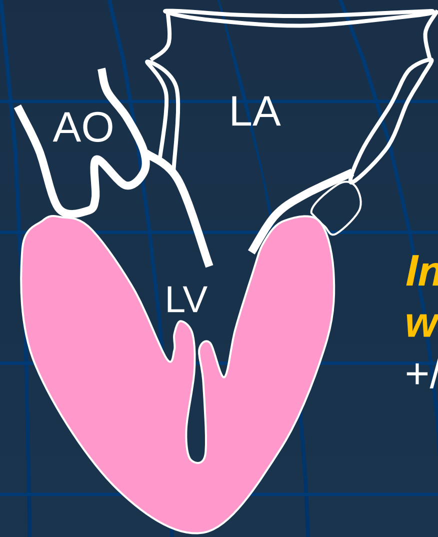
1



LV dilation
systolic dysfunction

Dilated cardiomyopathy

2



Inappropriate ↑LV
wall thickness
+/- LVOT obstruction

Hypertrophic cardiomyopathy

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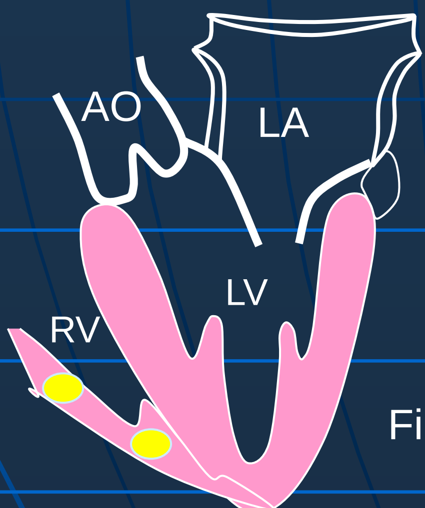


Fibro-adipose infiltrate

Arrhythmogenic right ventricular cardiomyopthy

4

↑Myocardial stiffness
Abnormal LV filling
Diastolic dysfunction

Restrictive cardiomyopathy

Dilated Cardiomyopahthy (DCM)

Infection

Autoimmune

Metabolic / endocrine dysfunction

Infiltrative

Drugs / alcohol / toxins

Pregnancy

Neuromuscular disorders

Dilated LV and reduced LVEF not explained by the severity of CAD, VHD, CHD or HT

Heart Failure

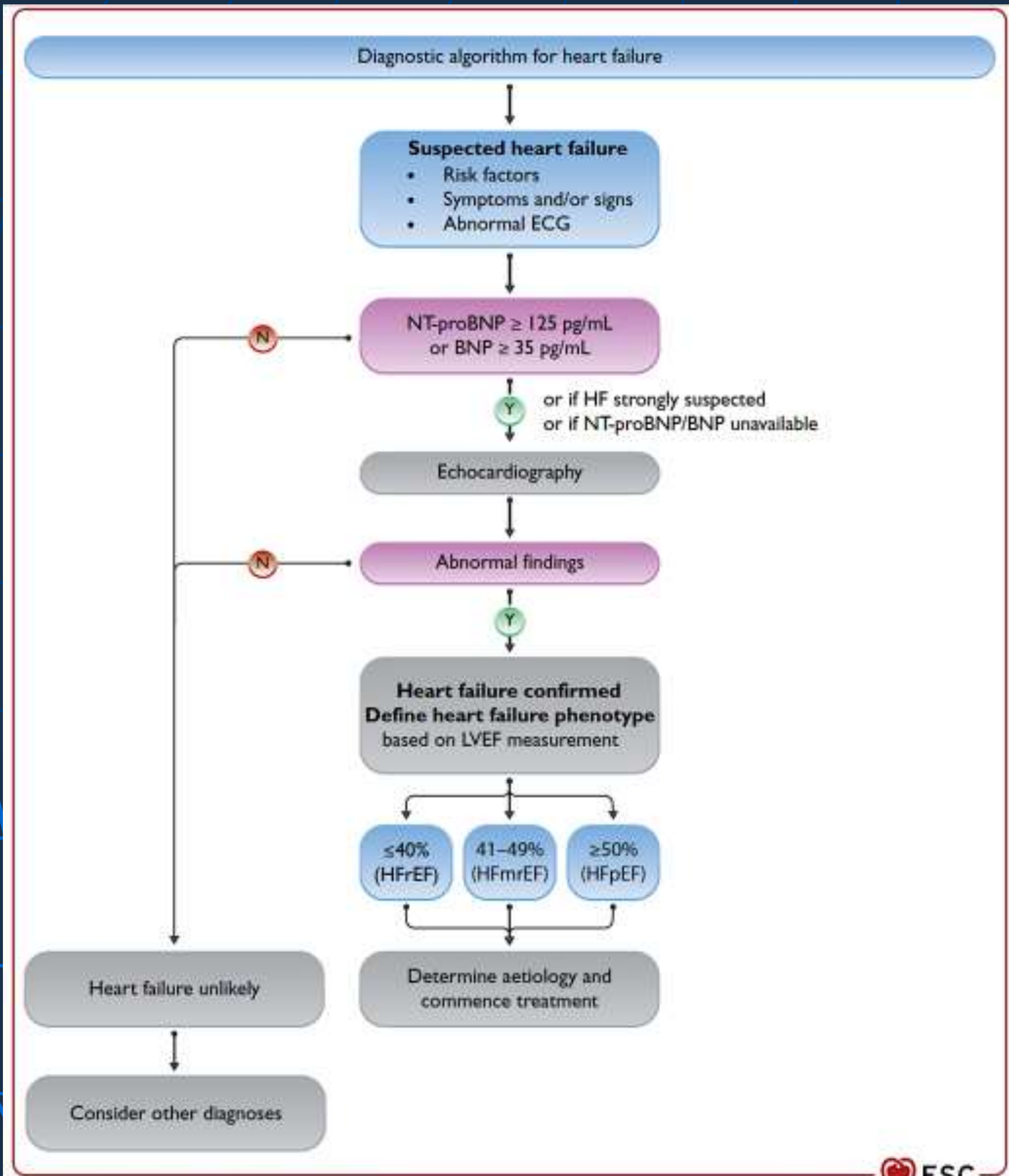
Sudden Cardiac Death

- **1:250 – 1:500 individuals**
- **~ 60% of all cardiomyopathies**
- **Clinically heterogeneous diseases**
- **Idiopathic DCM ~35%**
- **Positive familial history in ~50% of pts with idiopathic DCM**

Familial DCM

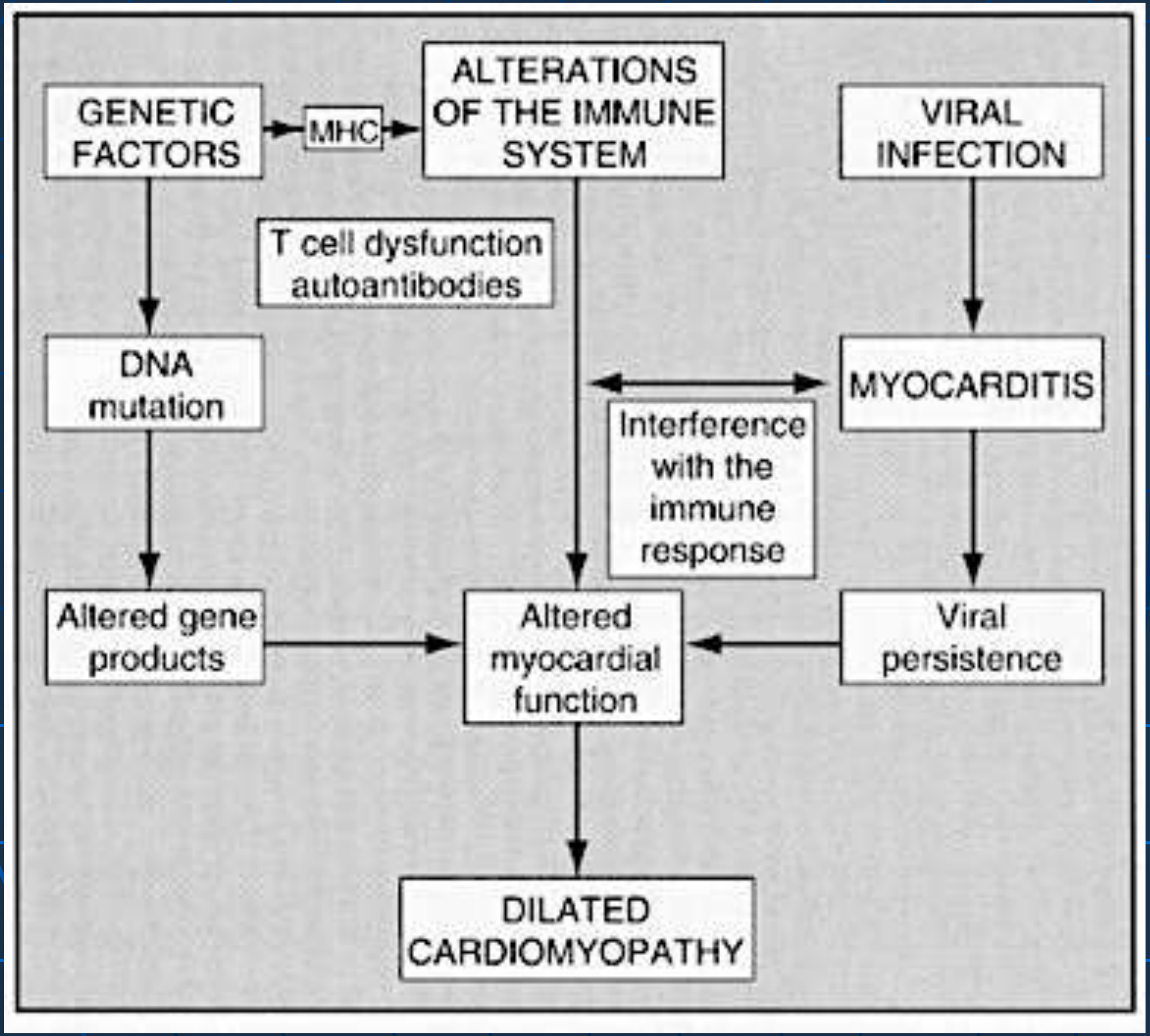
- 15-30% DCM
- Onset: 20-50 y/o
- ≥2 1st / 2nd degree relatives have DCM or a 1st degree relative has autopsy proven DCM and SCD <50 y/o
- Multiple genetic etiologies:
  - >90% autosomal dominant
  - Most common is Titin (TTN, 20-25%) and Lamin A/C (LMNA ~6%)
  - *Neuromuscular disorders e.g. Duchene muscular dystrophy, Becker muscular dystrophy, myotonic dystrophy; Mitochondrial diseases*
- Challenge in diagnosis of Familial DCMP:
  - Incomplete family history
  - Variable age-related penetrance
  - Concomitant cardiac conditions

Diagnostic Workup of DCM



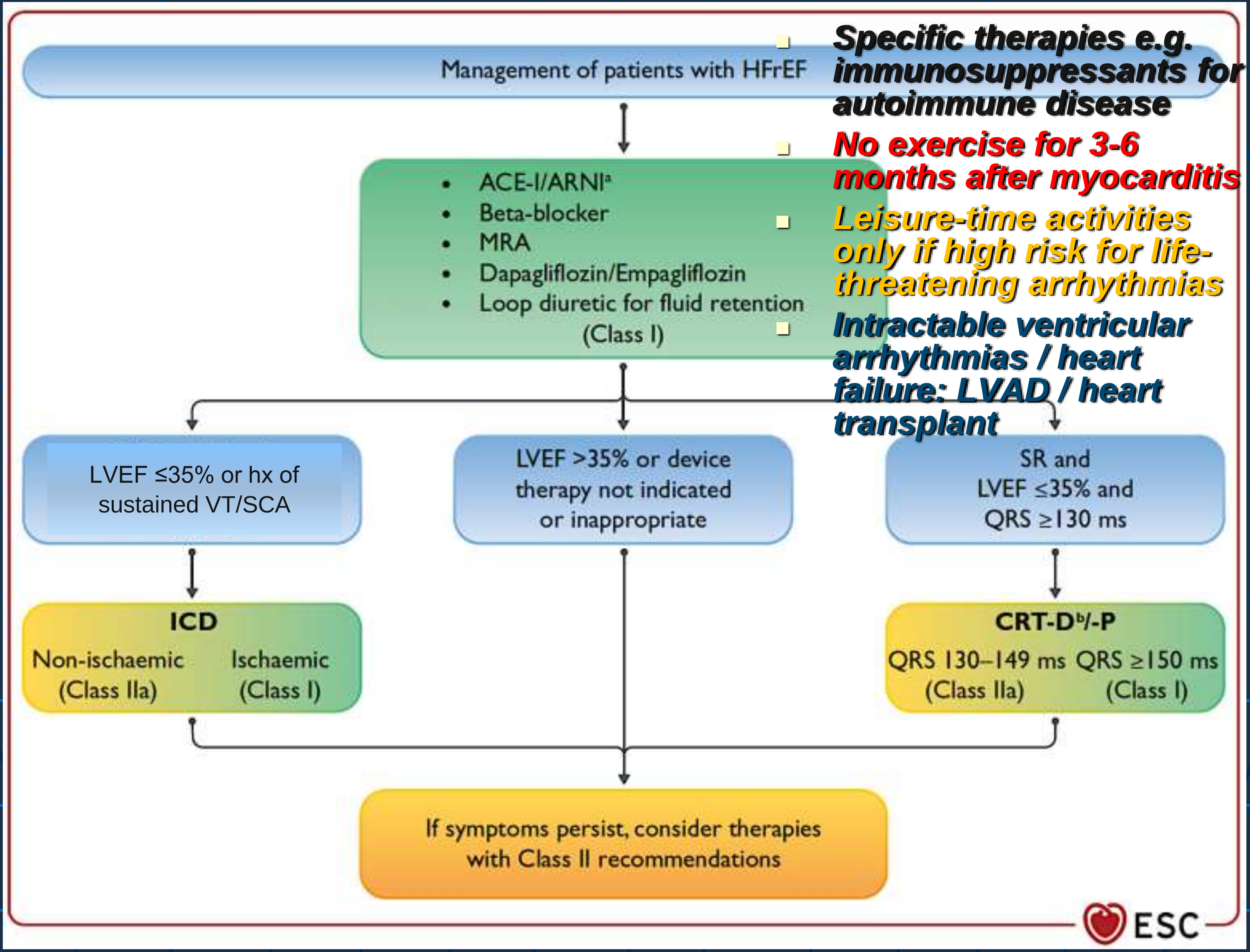
- LV end-diastolic volumes / diameter > 2SD from normal according to normograms corrected for age and BSA
- LVEF <50%
- ECG: no specific features but usually abnormal; AF, AVB, LBBB,
- Cardiac MRI e.g. LGE pattern; edema
- CT coronary angiogram
- FDG-PET scan for sarcoidosis
- Endomyocardial biopsy if suspected inflammatory / infiltrative CM
- Blood tests
- Idiopathic DCM – genetic counseling (3-generation family tree; genetic screening)

Pathogenesis of DCM



- **Enteroviruses esp coxsackie A/B**
- **Herpesvirus; adenoviruses; parvoviruses**

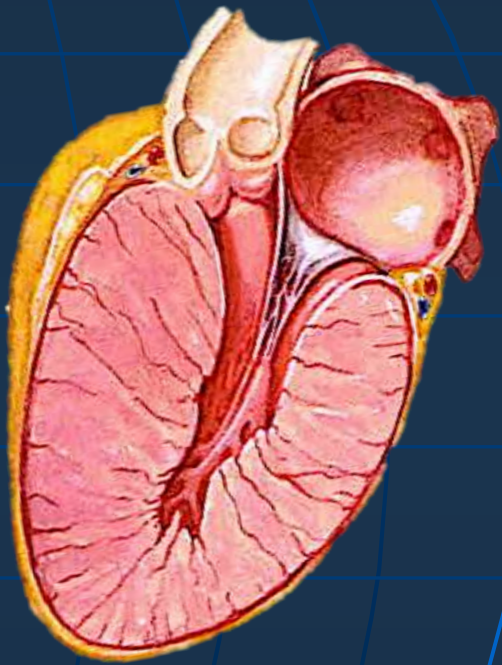
Management of DCM



- ***Specific therapies e.g. immunosuppressants for autoimmune disease***
- ***No exercise for 3-6 months after myocarditis***
- ***Leisure-time activities only if high risk for life-threatening arrhythmias***
- ***Intractable ventricular arrhythmias / heart failure: LVAD / heart transplant***

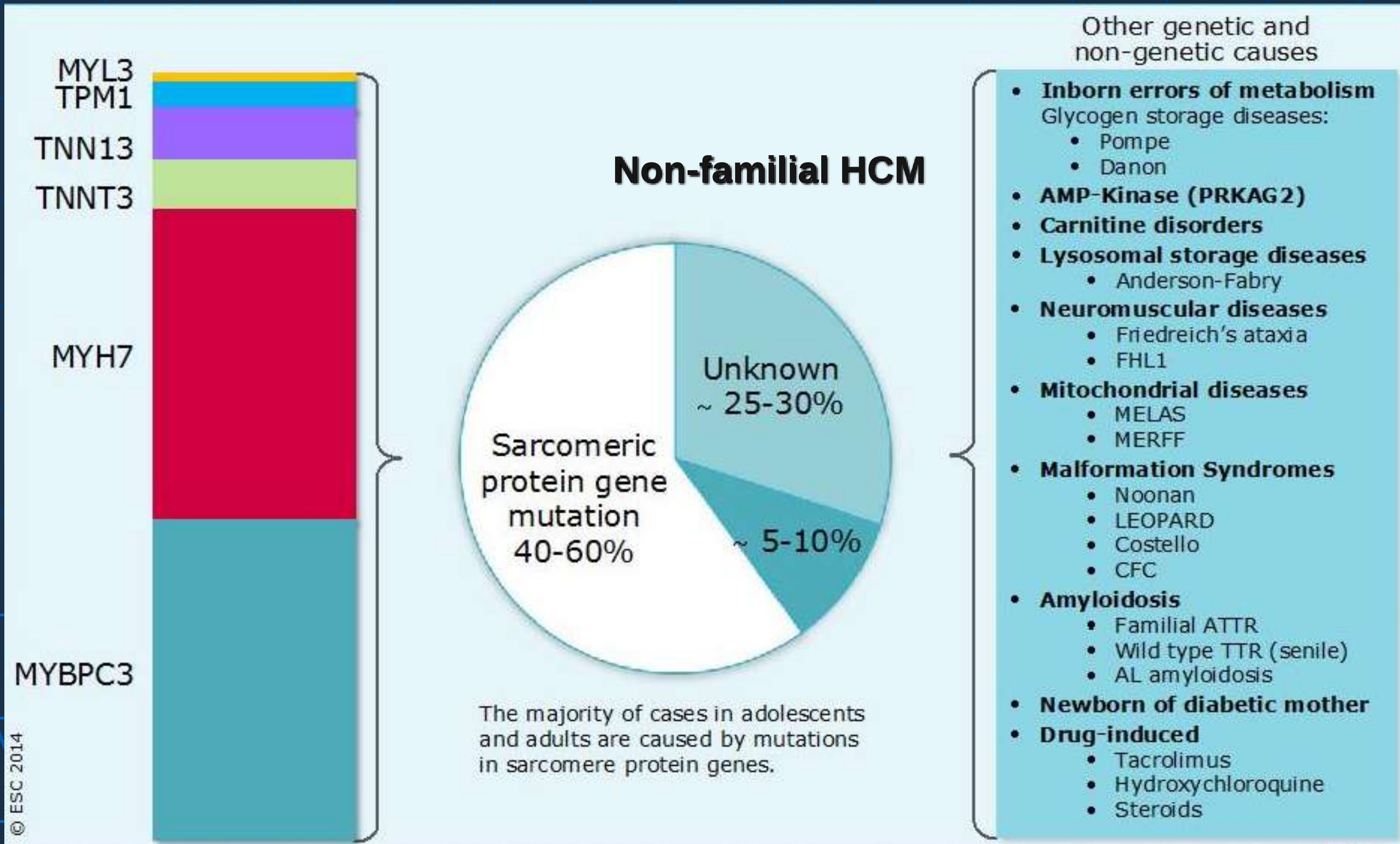
2020 ESC Guidelines for Heart Failure

Hypertrophic Cardiomyopathy (HCM)



- ↑left ventricular wall thickness at end-diastole that cannot be explained by abnormal loading condition
- 1:500 to 1:1000 individuals
- Major cause of premature sudden cardiac death in young and apparently healthy athletes
- Autosomal dominant inheritance
- Genetically heterogeneous diseases
- ≥2 1st / 2nd degree relatives with HCM / a 1st degree relative has autopsy proven HCM and SCD <50 y/o
- Variable penetrance, clinical presentation and prognosis

Diverse Etiologies of HCM



Clinical Presentations of HCM

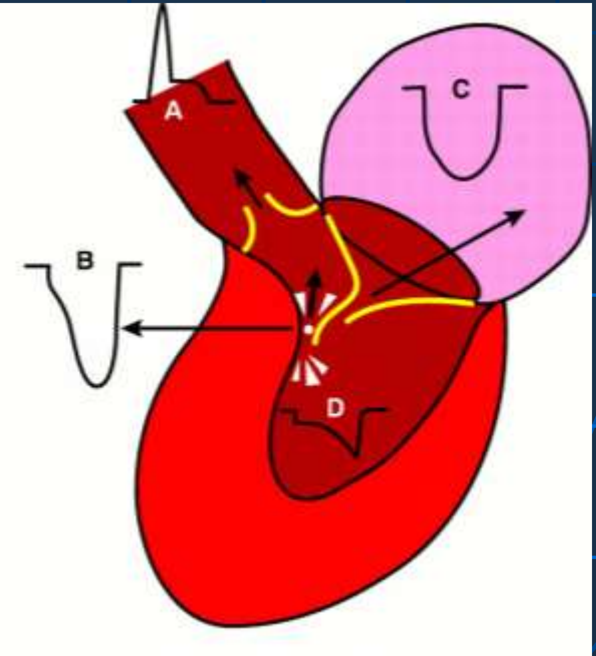
- Variable ages at onset and clinical presentations even with the same disease-causing mutation
- Variable degree and location of hypertrophy
- Incidental ECG/Echo findings
- 30-40% in referral-based cohorts will experience adverse events:
  - Dyspnea on exertion (90%), orthopnea, PND due to LVOTO during exertion, systolic HF or diastolic HF
  - Angina (70-80%) due to LVH ± LVOTO during exertion
  - Syncope (20%) / Pre-syncope (50%) due to LVOTO on exertion or cardiac arrhythmias (AT, AF, VT, VF)
  - Stroke and thromboembolism due to AF
  - Sudden cardiac death

Pathophysiology of Hypertrophic Obstructive Cardiomyopathy (HOCM)

4 interrelated processes

- LVOTO (asymmetrical septal hypertrophy and systolic anterior motion of AMVL)
- Diastolic dysfunction
- Myocardial ischemia
- Mitral regurgitation

Worse survival than those without obstruction



Mitral leaflet – septal contact

Diagnostic Workup of HCM

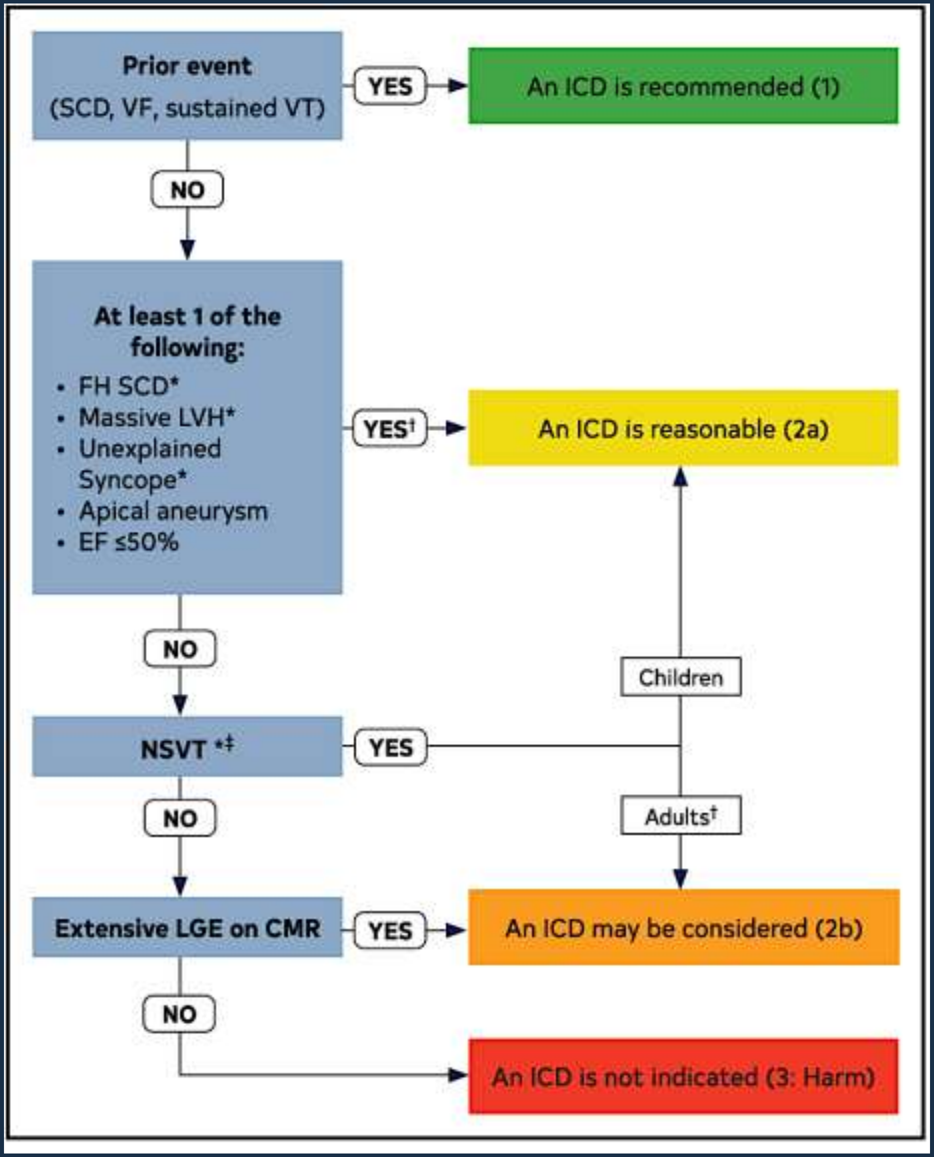
- Adults: ≥ 15mm or ≥ 13mm if FHx +ve / genetic test +ve
- Children: BSA-adjusted z-score ≥ 2.5SD above mean or ≥2SD above mean if FHx +ve / genetic test +ve
- Asymmetrical septal hypertrophy, symmetrical hypertrophy, apical hypertrophy
- ECG: LVH with strain (symmetrical or asymmetrical) / precordial TWI (apical) / low voltage (amyloidosis)
- 3-generation family history / genetic testing
- Echo: LVOTO (resting / provocated / exercise): ≥ 30mmHg (≥ 50 mmHg: require treatment)
- Holter: Non-sustained ventricular tachycardia
- Cardiac MRI: Extensive late gadolinium enhancement

Risk Factors of Sudden Cardiac Death (SCD) in HCM

Family history of sudden death from HCM	Sudden death judged definitively or likely attributable to HCM in ≥1 first-degree or close relatives who are ≤50 y of age. Close relatives would generally be second-degree relatives; however, multiple SCDs in tertiary relatives should also be considered relevant.
Massive LVH	Wall thickness ≥30 mm in any segment within the chamber by echocardiography or CMR imaging; consideration for this morphologic marker is also given to borderline values of ≥28 mm in individual patients at the discretion of the treating cardiologist. For pediatric patients with HCM, an absolute or z-score threshold for wall thickness has not been established; however, a maximal wall that corresponds to a z-score ≥20 (and >10 in conjunction with other risk factors) appears reasonable.
Unexplained syncope	≥1 Unexplained episodes involving acute transient loss of consciousness, judged by history unlikely to be of neurocardiogenic (vasovagal) etiology, nor attributable to LVOTO, and especially when occurring within 6 mo of evaluation (events beyond 5 y in the past do not appear to have relevance).
HCM with LV systolic dysfunction	Systolic dysfunction with EF <50% by echocardiography or CMR imaging.
LV apical aneurysm	Apical aneurysm defined as a discrete thin-walled dyskinetic or akinetic segment of the most distal portion of the LV chamber; independent of size.
Extensive LGE on CMR imaging	Diffuse and extensive LGE, representing fibrosis, either quantified or estimated by visual inspection, comprising ≥15% of LV mass (extent of LGE conferring risk has not been established in children).
NSVT on ambulatory monitor	It would seem most appropriate to place greater weight on NSVT as a risk marker when runs are frequent (≥3), longer (≥10 beats), and faster (≥200 bpm) occurring usually over 24 to 48 h of monitoring. For pediatric patients, a VT rate that exceeds the baseline sinus rate by >20% is considered significant.

2020 AHA/ACC Guidelines for Hypertrophic Cardiomyopathy

Flowchart for ICD implantation



Colors correspond to the Class of Recommendation in Table 2. *ICD decisions in pediatric patients with HCM are based on ≥1 of these major risk factors: family history of HCM SCD, NSVT on ambulatory monitor, massive LVH, and unexplained syncope. †In patients >16 years of age, 5-year risk estimates can be considered to fully inform patients during shared decision-making discussions. ‡It would seem most appropriate to place greater weight on frequent, longer, and faster runs of NSVT. CMR indicates cardiovascular magnetic resonance; EF, ejection fraction; FH, family history; HCM, hypertrophic cardiomyopathy; ICD, implantable cardioverter-defibrillator; LGE, late gadolinium enhancement; LVH, left ventricular hypertrophy; NSVT, nonsustained ventricular tachycardia; SCD, sudden cardiac death; VF, ventricular fibrillation; and VT, ventricular tachycardia.

2020 AHA/ACC Guidelines for Hypertrophic Cardiomyopathy

Management of HCM

- SCD risk assessment & stress testing at baseline
- Genetic counseling
- Evaluate clinical, echo, holter ± stress testing every 1-2 years or when symptoms change
- ***Exercise: Mild to moderate intensity recreational exercise is acceptable for most patients***
- Obstructive and symptomatic: Avoid vasodilators. Beta-blocker, non-dihydropyridine calcium channel blockers, disopyramide, septal ablation, myectomy
- Systolic dysfunction: Avoid negative inotropics. ARNI / ACEI / ARB, BB, MRA, CRT implantation as per HF guidelines
- High SCD risk: ICD implantation
- Recurrent ventricular arrhythmias: antiarrhythmics
- Intractable ventricular arrhythmias / heart failure: LVAD / heart transplant

2020 AHA/ACC Guidelines for Hypertrophic Cardiomyopathy

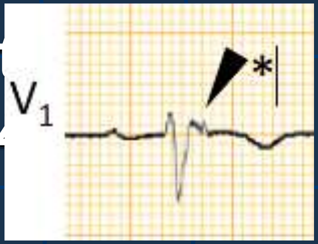
Arrhythmogenic Right Ventricular Cardiomyopathy (ARVC)

- **Arrhythmogenic cardiomyopathy: arrhythmogenic myocardial disorder not explained by CAD, VHD, CHD or HT**
- **ARVC: predominant RV involvement with fibrous / fibrofatty replacement of RV myocardium**
- **High prevalence in some region in Italy**
- **Most commonly autosomal dominant but variable penetrance and expressivity, 1/3 familial. mostly involve genes encoding desmosome, intercalated disc and ion channels**
- **Ventricular arrhythmias with LBBB morphology / SCD in the young**
- **Disease progression related to exercise**

From Marcus F 2000

Summary of the Modified Task Force Criteria for ARVC

- **Global / regional dysfunction or structural alterations:** Echo / MRI / RV angiography showing RV akinesia, dyskinesia, aneurysms and significant dilatation / reduced systolic function
- **ECG:** Depolarization (Epsilon wave) or repolarizat[...] wave inversion at V1-V3 or beyond in patients >1[...] without complete RBBB) abnormalities



- **Arrhythmias:** VT / NSVT of LBBB morphology and superior axis
- **Family history:** ≥ one 1st degree relatives with ARVC diagnosed pathologically / by Task Force Criteria
- **Tissue characterization:** Endomyocardial biopsy showing significant fibrous replacement of RV free wall myocardium

Risk Factors of SCD in ARVC

- **Hx of VT / SCA**
- **LVEF ≤ 35%**
- **Specific major risk factors**
  - **NSVT, inducible VT at electrophysiology study, LVEF ≤ 49%**
- **Specific minor risk factors**
  - **Male, >1000 PVCs / 24 hours, RV dysfunction, proband status, ≥ 2 desmosomal variants**
- ***3 major, 2 major + 2 minor or 1 major + 4 minor risk factors***
- *If NSVT is present, PVC criteria is not counted

2019 HRS Expert Consensus Statement on ARVC

Management of ARVC

- **Genetic counseling**
- **Management of arrhythmias:**
  - **Consideration of ICD; betablockers if no ICD; antiarrhythmics and betablockers if recurrent ventricular arrhythmias**
- **Management of RV dysfunction:**
  - **Symptomatic RV dysfunction: nitrate to reduce preload; may try ACEI / ARB / BB / AA / diuretics**
- **Management of thromboembolic risk:**
  - **Antithrombotic therapy if AF, intracavitary thrombosis, thromboembolism**
- ***Exercise: No competitive or frequent high-intensity endurance exercise that is associated with ↑ventricular arrhythmias and progression of disease***
- ***Intractable ventricular arrhythmias / heart failure: LVAD / heart transplant***

2019 HRS Expert Consensus Statement on ARVC

Restrictive Cardiomyopathy (RCM)

- ↑Myocardial stiffness leading to an extremely impaired ventricular filling and diastolic dysfunction called ‘restrictive physiology’
- Presents as HFrEF but arrhythmias possible
- The least common form of CM
- Mostly acquired
- Idiopathic: relentless symptomatic progression and high mortality
- Secondary to systemic diseases
- Autosomal dominant (mutations in sarcomere subunits)

Restrictive Cardiomyopathy (RCM)

- Functional resemblance to constrictive pericarditis. Differentiating the two is mandatory because of the potential for successful surgical treatment of constriction
- Echocardiogram
- Right heart catheterization
- Cardiac MRI
- Nuclear imaging (technetium-labelled bisphosphonates for ATTR-cardiac amyloidosis; gallium-67 scan for sarcoidosis)
- FDG-PET imaging for sarcoidosis
- Endomyocardial biopsy
- Blood tests; other biopsies
- Treatment may be available for specific conditions e.g. enzyme replacement for Fabry disease, tafamidis for TTR-cardiac amyloidosis

Etiologies of RCM

Infiltrative		
Amyloidosis	Acquired/ inherited	*TTR* gene variants (V122I; I68L; L111M; T60A; S23N; P24S; W41L; V30M; V20I), *APOA1*
Sarcoidosis	Acquired	
Primary hyperoxaluria	Inherited	*AGXT (type 1), GRHPR (type 2), HOGA1 (type 3)*
Storage diseases		
Fabry disease	Inherited	*GLA*
Gaucher disease	Inherited	*GBA*
Hereditary hemochromatosis	Inherited	*HAMP, HFE, HFE2, HJV, PNPLA3, SLC40A1, TfR2*
Glycogen storage disease	Inherited	Per specific type
Mucopolysaccharidosis type I (Hurler syndrome)	Inherited	*IDUA*
Mucopolysaccharidosis type II (Hunter syndrome)	Inherited	IDS
Niemann–Pick disease	Inherited	NPC1, NPC2, SMPD1

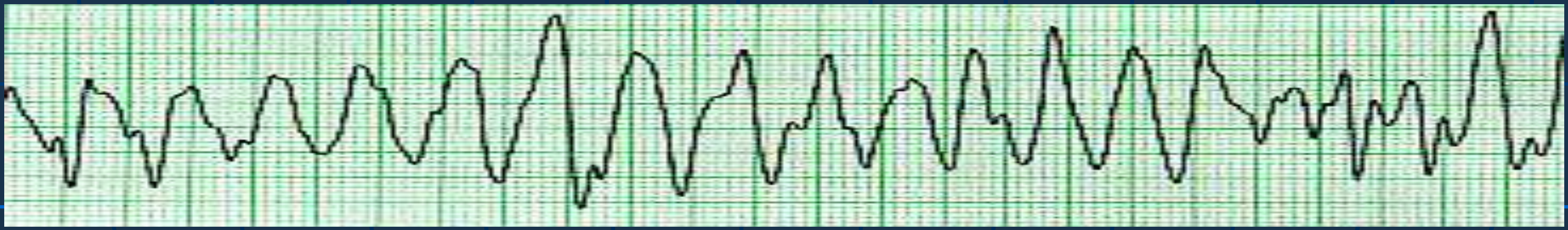
Etiologies of RCM

Noninfiltrative		
Idiopathic	Acquired	
Diabetic cardiomyopathy	Acquired	
Scleroderma	Acquired	
Myofibrillar myopathies	Inherited	*BAG3, CRYAB, DES, DNAJB6, FHL1, FLNC, LDB3, MYOT*
Pseuxanthoma elasticum	Inherited	ABCC6
Sarcomeric protein disorders	Inherited	*ACTC, β-MHC, TNNT2, TNNI3, TNNC1, DES, MYH, MYL3, CRYAB*
Werner's syndrome	Inherited	*WRN*

Etiologies of RCM

Endomyocardial		
Carcinoid heart disease	Acquired	
Endomyocardial fibrosis		
Idiopathic	Acquired	
Hypereosinophilic syndrome	Acquired	
Chronic eosinophilic leukemia	Acquired	
Drugs (serotonin, methysergide, ergotamine, mercurial agents, busulfan)	Acquired	
Endocardial fibroelastosis	Inherited	*BMP5, BMP7, TAZ*
Consequence of cancer/ cancer therapy		
Metastatic cancer	Acquired	
Drugs (anthracyclines)	Acquired	
Radiation	Acquired	

Primary Disorders of Rhythm and Conduction



Long QT syndrome (LQTS)

- LQTS is a disorder characterized by
  - (1) QT prolongation on ECG
  - (2) a propensity to ventricular tachyarrhythmias
- Usual presentations: syncope, seizures or SCD
- Can remain asymptomatic
- Ion channelopathy
- Congenital: Autosomal dominant; autosomal recessive; can have extracardiac involvement e.g. Andersen syndrome (LQT7), Timothy syndrome (LQT8)
- Acquired

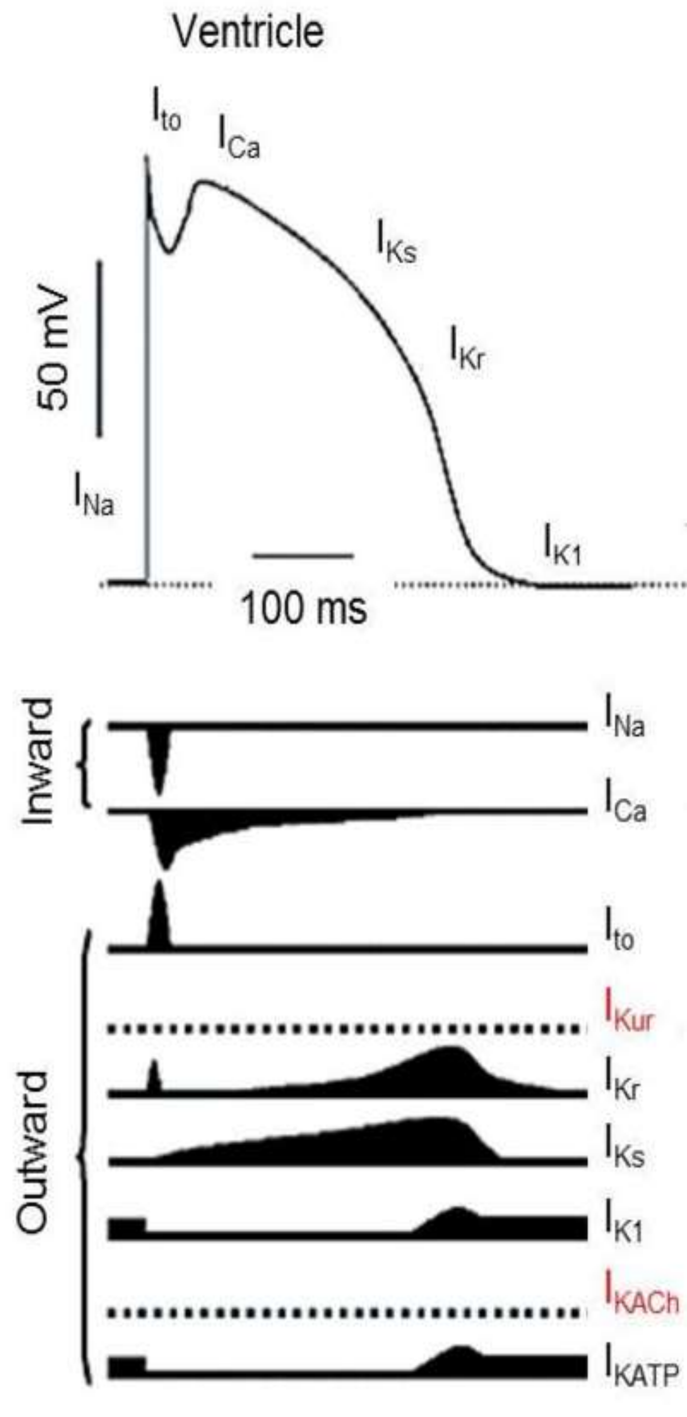
Definition of QT prolongation

- QT interval: (the beginning of QRS complex to the end of the T wave)
- Correct QT for HR e.g. the Bazett equation QTc = QT/√RR

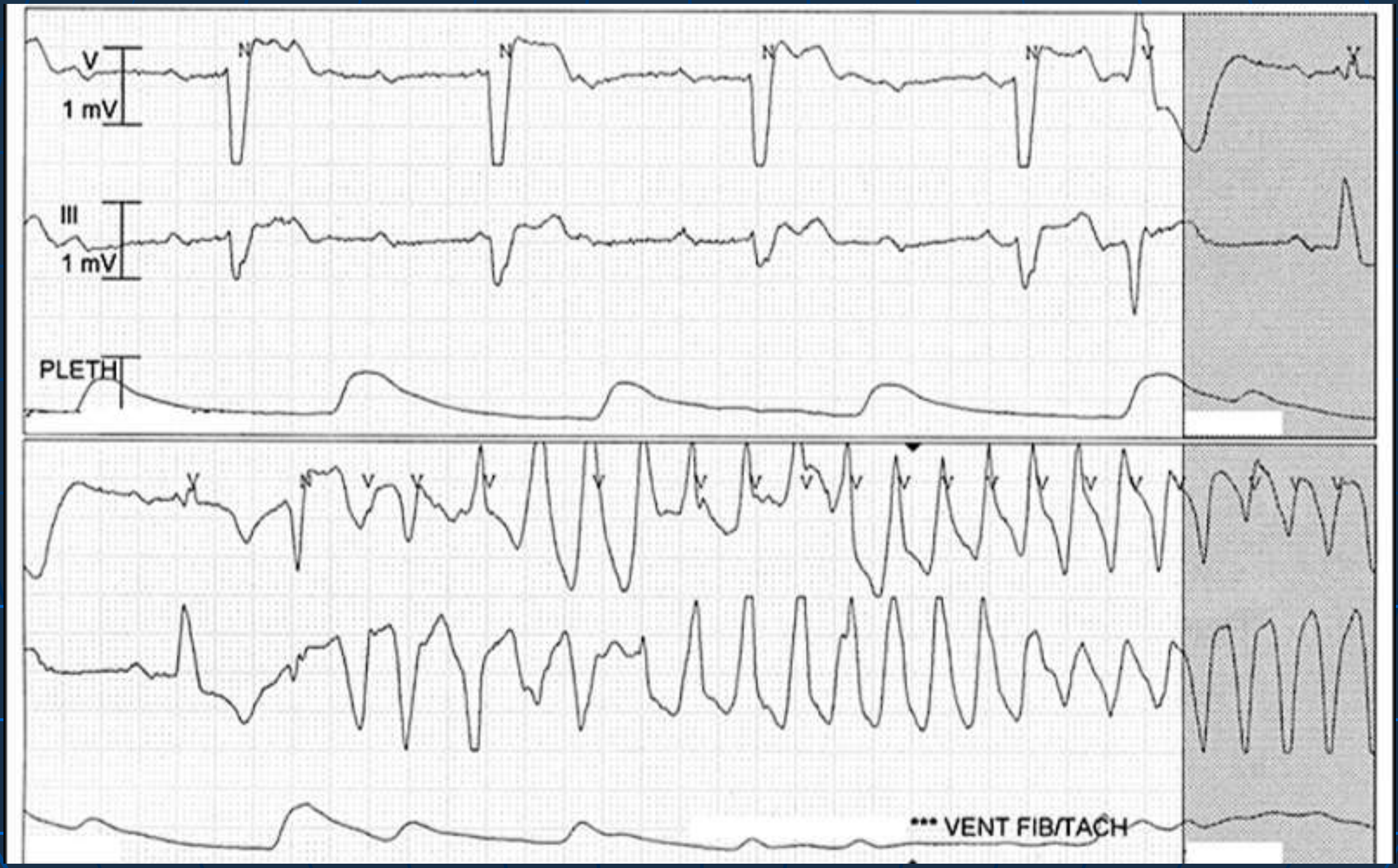
Age and Sex	Prolonged QTc (Seconds)	Borderline QTc (Seconds)	Reference Range (Seconds)
Children (<15 y)	>0.46	0.44-0.46	<0.44
Adult males	>0.45	0.43-0.45	<0.43
Adult females	>0.46	0.45-0.46	<0.45

Pathophysiology

- The QT interval represents ventricular repolarization
- Prolonged repolarization secondary to
  - Reduced repolarization current e.g. I_{ks} & I_{Kr}
  - Enhanced depolarization current e.g. I_{Na}
- Functional substrate for transmural reentry, causing polymorphic ventricular tachycardia, or torsade de pointes (TdP)



Pause-dependent torsade-de-pointes in Acquired LQTS



Kearney KE et al. AJM 2015

Congenital LQTS

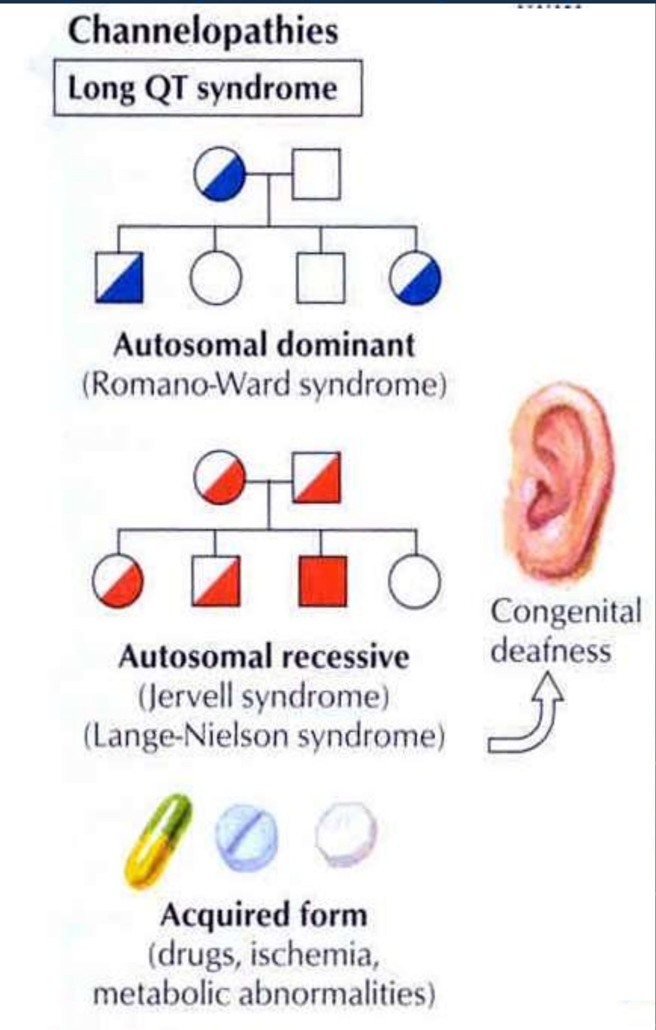


TABLE 1. Genes Envolved in the Long QT Syndrome

Type	Locus	Gene	Protein
Romano-Ward (autosomal dominant)			
LQTS1	11p15.5	*KCNQ1/KVLQT1*	Principal, I_{Ks} α-subunit
LQTS2	7q35-36	*KCNH2/HERG*	Principal, I_{Kr} α-subunit
LQTS3	3p21-p24	*SCN5A*	Principal, I_{Na} α-subunit
LQTS4	4q25-q27	*ANKB*	Accessory, ankyrin-β
LQTS5	21q22.1	*KCNE1/minK*	Accessory, I_{Ks} β-subunit
LQTS6	21q22.1	*KCNE2/MiRP1*	Accessory, I_{Kr} β-subunit
LQTS7[a]	17q23	*KCNJ8*	Principal, $K_{ir}2.1$ α-subunit
LQTS8[b]	12p13.3	*CACNA1*	Principal, $C_{av}1.2$ α-subunit
LQTS9	3p25	*CAV3*	Accessory, caveolin 3
LQTS10	11q23	*SCN4B*	Accessory, I_{Na} β4-subunit
Jervell-Lange-Nielsen (autosomal recessive)			
JLN1	11p15.5	KCNQ1/KVLQT1	Principal, I_{Ks} α-subunit
JLN	21q22.1	KCNE1/minK	Accessory, I_{Ks} β-subunit

[a]Andersen-Tawil Syndrome
[b]Timothy syndrome

Medeiros-Domingo A et al. Revista Espanola De Cardiologia 2007

Type	Current	Functional Effect	Frequency Among LQTS	ECG[12,13]	Triggers Lethal Cardiac Event[10]	Penetrance*
LQTS1	K	↓	30%-35%		Exercise (68%) Emotional Stress (14%) Sleep, Repose (9%) Others (19%)	62%
LQTS2	K	↓	25%-30%		Exercise (29%) Emotional Stress (49%) Sleep, Repose (22%)	75%
LQTS3	Na	↑	5%-10%		Exercise (4%) Emotional Stress (12%) Sleep, Repose (64%) Others (20%)	90%

Medical Conditions Associated with LQTS and Torsade de Pointes

Electrolyte imbalance

- Hypokalemia
- Hypocalcemia
- Hypomagnesemia

Medical Conditions

- Arrhythmias: AV block, severe sinus bradycardia and pauses
- Endocrine: DM, hyperparathyroidism, hypothyroidism and pheochromocytoma
- Neurologic: CVA, trauma, subarachnoid hemorrhage, encephalitis
- Nutritional: acute weight loss, alcoholism, liquid protein diet, starvation and obesity

Drugs Associated with LQTS and Torsade de Pointes

Antianginal
- Bepridil, Israpidine, Nicardipine, ranolazine

Antiarrhythmic Drugs
- *Class IA*
  - Quinidine, Procainamide
  - Disopyramide
- *Class III*
  - Sotalol, dronedarone
  - Ibutilide, dofetilide

Antibiotics
- Erythromycin, Trimethoprim & Sulfamethaxazole, Pentamidine,
- Clarithromycin, Moxifloxacin

Antihistamines
- Terfenedine, Astemizole, diphenhydramine

Muscle Relaxant
- Tizanidine

Antifungal Agents
- Ketoconazole
- Flucoconazole
- Itraconazole

Diuretics
- Indapamide

Gastrointestinal
- Cisapride

Lipid Lowering
- Probucol

Psychotropics
- Phenothiazines, Tricyclic antidepressants (Amitriptyline)
- Haloperidol, Risperidone, Pimozide

Immunosuppressives
- Tacrolimus

Sedative/Hypnotics
- Chloral hydrate

Toxins
- Anthopleurin-A, ATX-II
- Organophosphate insecticides
- Liquid protein diets

Diagnosis of LQTS

- **Clinical Syndrome Recognition**
  - **Symptoms**
  - **Family history**
  - **Other ECG findings**
  - **Arrhythmias**
- ***Provocative Testing:***

  - exercise stress test; epinephrine, isoproterenol

  - maladaptation of the QT interval to the changing heart rate, with evident QTc prolongation ± VT
- ***Exclusion of reversible causes:***

  - electrolyte, thyroid function test
- ***Genetic Identification:***

  - Identification of an LQTS gene mutation confirms the diagnosis

  - A negative genetic testing has limited diagnostic value as it could be -ve in 25% to 50%

Schwartz Score

Table 1 Diagnostic criteria LQTS—1993 Schwartz[6]

ECG findings[a]	
(A) QTc[b]	
≥ 480 ms	3 points
460–479 ms	2 points
450–459 ms (in males)	1 point
(B) Torsade de pointes[c]	2 points
(C) T wave alternans	1 point
(D) Notched T wave in three leads	1 point
(E) Low heart rate for age[d]	0.5 point
Clinical history	
(A) Syncope[c]	
With stress	2 points
Without stress	1 point
(B) Congenital deafness	0.5 point
Family history[e]	
(A) Family members with definite LQTS[f]	1 point
(B) Unexplained sudden cardiac death below age 30 among immediate family members	0.5 point

Definite LQTS: >/=4
High probability: >/=3.5
Intermediate probability: 1.5-3
Low probability </=1

a: no medications / disorders accountable for the changes
b: Bazett's formula
c: Mutually exclusive
d: resting heart rate below second centile for age
e: Same family member cannot be counted in both A & B

Management of LQTS

- **Genetic Counseling**
- **Beta-blockers (adrenergic blockade) are effective in preventing cardiac events in approximately 70% of patients**
- **Life-style advices**
  - ***Exercise advices* – Asymptomatic (± BB) : competitive exercise exercise ; symptomatic but not high risk: low to moderate intensity exercise; specific exercise advices based on genetic mutation; avoidance of electrolyte imbalance, heatstroke, and use of AED**
  - ***Medication advices***
- **ICD if SCA or symptomatic despite betablockers**

2017 AHA/ACC/HRS Guideline for VA and SCD

Management of LQTS

- **Cardiac pacemakers:** eliminates arrhythmogenic bradycardia, decreases heart rate irregularities (eliminating short-long-short series), and decreases repolarization heterogeneity, therefore diminishing the risk of torsade de pointes ventricular tachycardia.
- **Left cervicothoracic stellectomy** is another anti-adrenergic therapeutic measure used in high-risk patients with LQTS, especially in those with recurrences of cardiac events despite beta-blocker therapy
- In some high-risk patients, **combination therapy** consisting of beta-blockers, stellectomy, and implantation of cardioverter-defibrillator with cardiac pacing function is used

2017 AHA/ACC/HRS Guideline for VA and SCD

Prognosis of LQTS

- Episodes of torsades de pointes usually are self-terminating in patients with LQTS; only approximately 4-5% of cardiac events are fatal
- Prognosis in patients with LQTS treated with beta-blockers is good overall
- Prognosis after implantation of cardioverter-defibrillators is very good
- Genotype +ve / phenotype –ve patients have a 10% risk of major cardiac events by 40 y/o if untreated

The Brugada Syndrome (BrS)

- Brugada Syndrome is a disorder characterized by

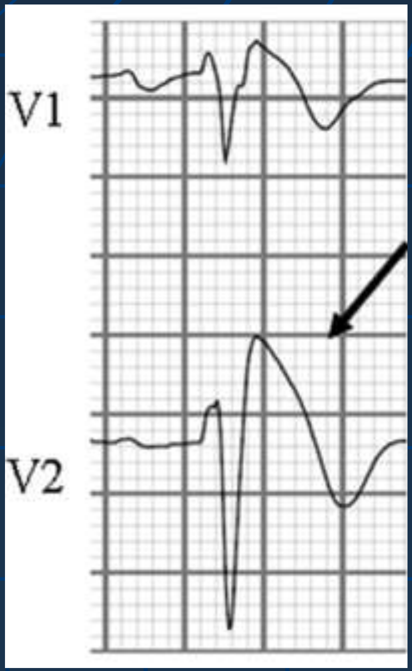
(1) ST segment elevation in the right-sided precordial leads

(2) A heart that is grossly structurally normal

(3) Conduction block at right ventricle

(4) Propensity for life-threatening ventricular tachyarrhythmias

- Mean age: mid-late 30's (4-70 yrs)
- M:F 8-10:1
- Associated with an ion channel gene mutation (Cardiac sodium channel gene SCN5A) resulting in loss of function resulting in ↓inward sodium current
- Autosomal dominant with variable penetrance
- Patients often show periodic normalization of ECG
- Manifest during fever / use of Na channel blockers



Type 1 ECG (coved-type ST-segment elevation)

The only diagnostic ECG in Brugada syndrome

- J-wave amplitude or an ST-segment elevation of ≥2 mm or 0.2 mV at its peak
- A negative T wave with little or no isoelectric separation

Type 2 ECG (saddle-back-type ST-segment elevation)

J-wave amplitude of ≥2 mm, gives rise to a gradually descending ST-segment elevation (remaining ≥1 mm above the baseline)

A positive or biphasic T wave that results in a saddle-back configuration.

Type 3 ECG

A right precordial ST-segment elevation (saddle-back type, coved type, or both) without meeting the aforementioned criteria

Yuka Mizusawa, and Arthur A.M. Wilde Circ Arrhythm Electrophysiol 2012

Clinical Presentations of BrS

- **In the majority of cases, malignant tachyarrhythmias occur at rest and at night (vagal influence)**
- **SCD due to VF is the first and only clinical event in some patients and arrhythmias recurrence rate is 40% over a 2-3-year follow-up**
- ***LQT/Brugada overlap syndrome***
- ***Possible overlap with arrhythmogenic right ventricular dysplasia (ARVD)***

Diagnosis of BrS

- Type I ECG pattern
- Latent or intermittent Brugada
  - Repeat ECG periodically
  - Repeat ECG at 2nd or 3rd ICS
  - Administration of ajmaline, procainamide or flecainide
- Brugada like ECG
  - RBBB, septal hypertrophy, ARVC
- Brugada ECG phenocopies
  - Myocardial ischemia; pulmonary embolism; myocardial and pericardial diseases; metabolic disorders; mechanical compression of RVOT

Management of BrS

- Genetic counseling
- Symptomatic individuals: ICD
  - E.g. syncope, seizure, nocturnal agonal respiration
- Otherwise: Follow-up (annual incidence of arrhythmic events = 0.5-1.2% per year)
- Isoproterenol / epinephrine / quinidine are useful for recurrent ventricular arrhythmias
- Life style advices
  - Avoid Brugada-aggravating drugs
  - Treat fever
  - Avoid excess alcohol
  - Avoid cocaine

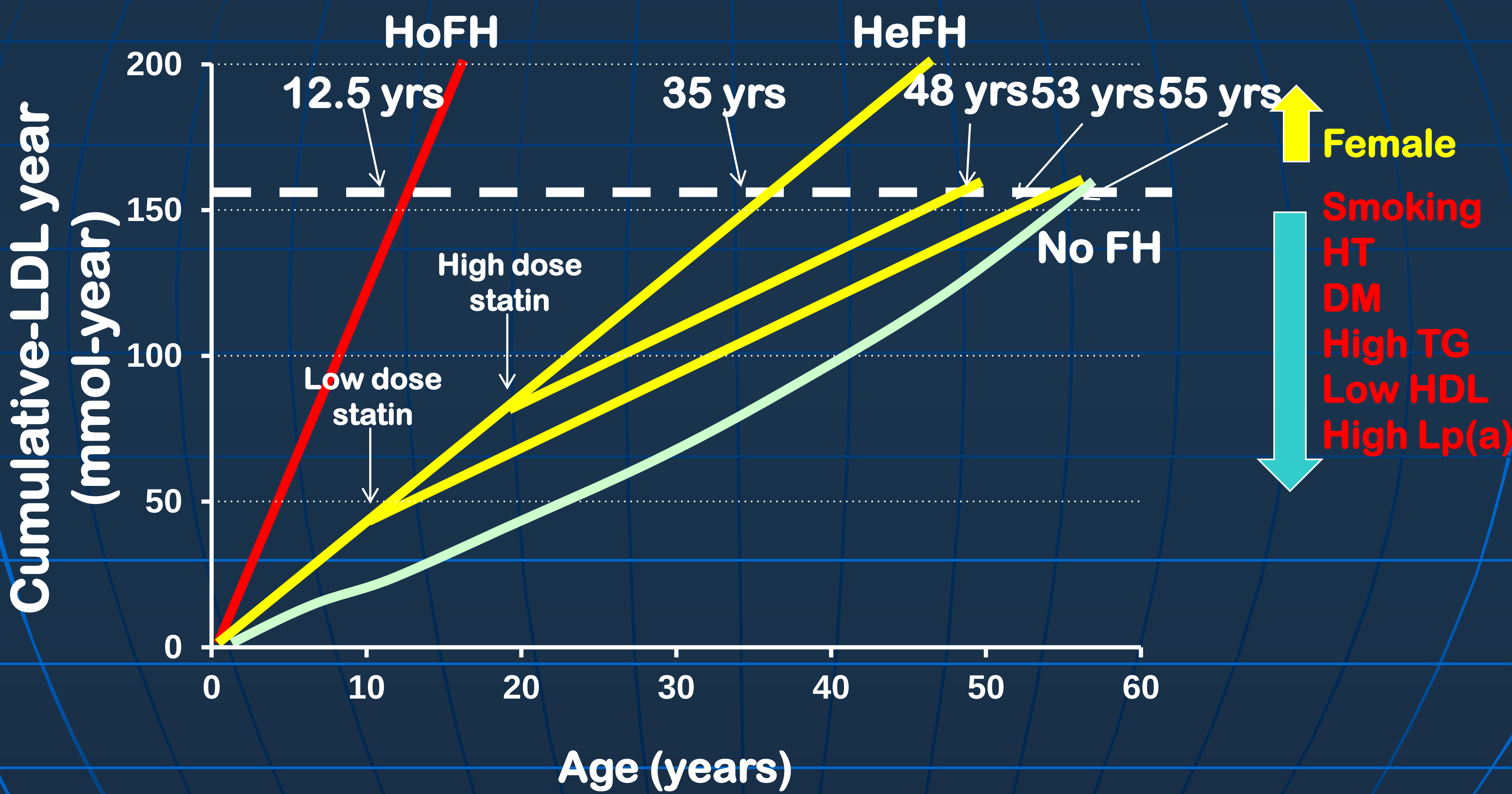
❖ ***Catheter ablation is the treatment of choice for refractory ventricular arrhythmias in many cardiomyopathies / channelopathies***

2017 AHA/ACC/HRS Guideline for VA and SCD

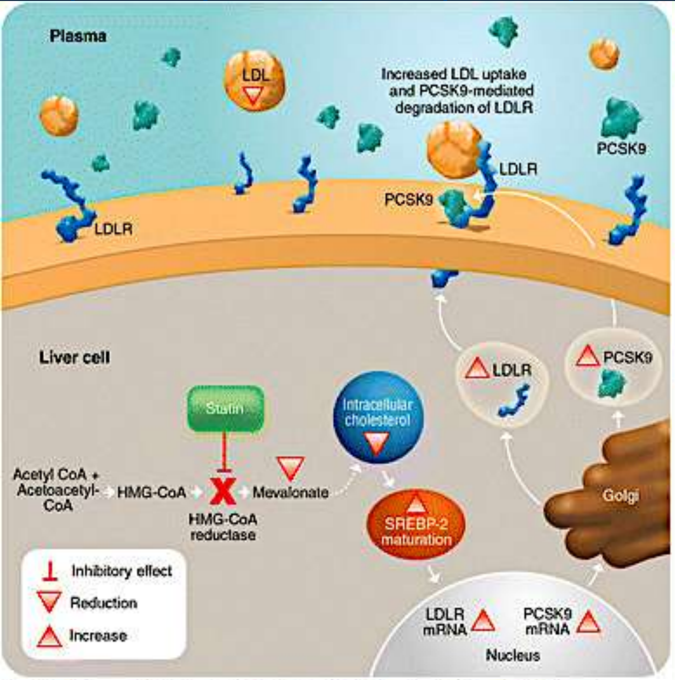
Familial Hypercholesterolemia (FH)

- FH is an autosomal dominant genetic disease, characterized by elevated LDL
- Heterozygous FH is around 1:200 to 1:500 individuals, mostly caused by mutations of LDLR, APOB, PCSK9, LDLPAR
- Increased risk of Premature atherosclerotic coronary heart disease
- Although current oral treatments (e.g., statins +/- ezetimibe, bile acid sequestrants and/or niacin) can reduce LDL-C of 50% – 65%, many patients are still unable to achieve recommended LDL targets

Cumulative LDL Burden In HoFH, HeFH And Non-FH

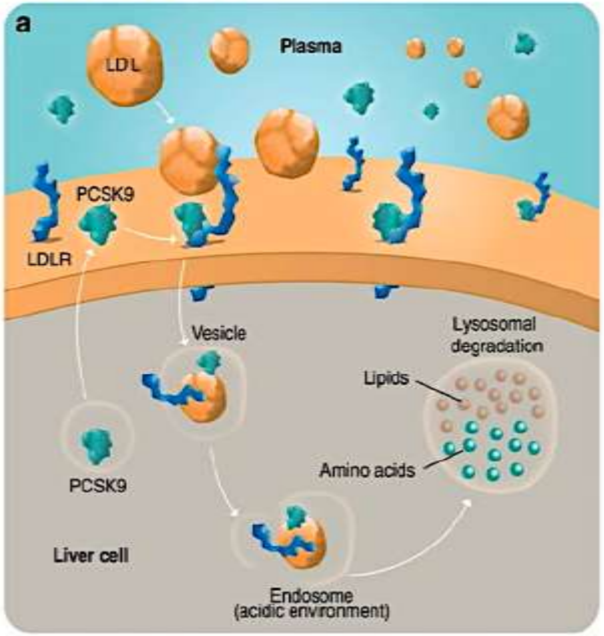


Modified Børge G. Nordestgaard EHJ 2013



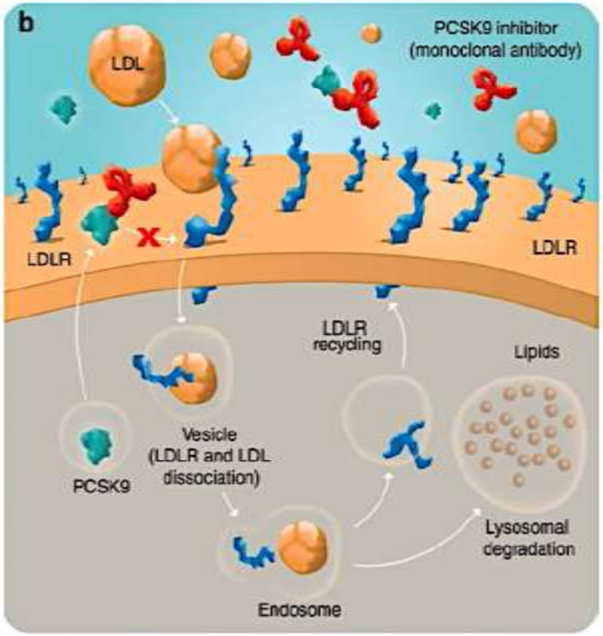
Impact of statins on cholesterol metabolism. Statins inhibit the biosynthesis of intracellular cholesterol by inhibiting HMG-CoA reductase. This results in low levels of intracellular cholesterol, leading to increased SREBP-2 activity, which promotes the production of PCSK9 and LDLR, the degradation of which is mediated by PCSK9 [72]. HMG CoA 3-hydroxy-3-methyl-glutaryl-CoA, LDL low-density lipoprotein, LDLR low density lipoprotein receptor, mRNA messenger RNA, PCSK9 proprotein convertase subtilisin/kexin type 9, SREBP-2 sterol-regulatory element-binding protein

How does PCSK9 work?



a) Secreted PCSK9 binds to LDLR on the liver cell surface and mediates the lysosomal degradation of the complex formed by PCSK9 - LDLR - LDL.

How does Inhibitors work?



b) In the presence of a monoclonal antibody that binds to PCSK9, the PCSK9-mediated degradation of LDLR is inhibited, resulting in an increased uptake of LDL-cholesterol by LDLR as more LDLR are recycled at the cell surface.

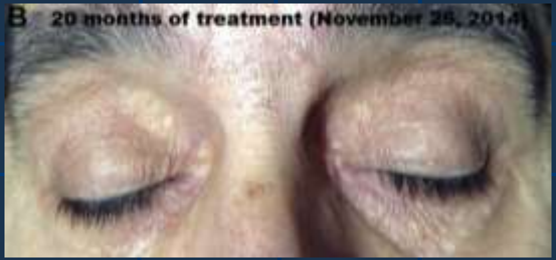
Krahenbuhl S et al. Drugs 2016.

Rapid Resolution Of Xanthelesma After Treatment With PCSK9 inhibitor: A Case report from ODYSSEY FH I

A 50-year-old man with HeFH (LDL mutation) diagnosed at 19 years with LDL of 7.76 mmol/L

Baseline: Atorvastatin 80 mg + Ezetimibe 10mg

LDL 4.24 mmol/L



20 months: Atorvastatin 80 mg + Ezetimibe 10mg + Alirocumab 75mg Q2w

LDL 0.59 mmol/L

26 months: Atorvastatin 80 mg + Ezetimibe 10mg + Alirocumab 75mg Q2w

LDL 1.21 mmol/L

Civeira F. J Clin Lipidol. 2016 Sep-Oct;10(5):1259-61.

Diagnosis and Management of FH

Whom to screen: index person or family member with
- Familial hypercholesterolaemia (FH)
- Cholesterol ≥8mmol/L (≥310mg/dL) for an adult (or) >95th percentile by age and gender for country
- Cholesterol ≥6mmol/L (≥230mg/dL) for a child (or) >95th percentile by age and gender for country
- Premature coronary heart disease
- Tendon xanthomas
- Sudden premature cardiac death in a family member

Diagnostic and treatment summary: Familial hypercholesterolaemia

Children:
No FHx of premature CAD:
LDL ≥ 5 mmol/L
FHx of premature CAD:
LDL ≥ 4 mmol/L
FHx of mutation:
LDL ≥3.6 mmol/L

↓

Diagnosis: Use the Dutch Lipid Clinic Network criteria (Table 1). This cannot be used in children.

↓

Exclude other causes of hypercholesterolaemia
Risk assessment: evaluate other cardiovascular risk factors, including elevated Lp(a)

→ Optional: Screen for asymptomatic atherosclerosis

↓

Cascade screen family using LDL cholesterol levels (draw pedigree as in *Figure 7*)

↓

If Dutch Lipid Clinic Network criteria (Table 1) score > 5
- Screen for causative mutation in index case (if DNA test is available in country)
- Followed by genetic testing of family if causative mutation is found (genetic cascade screening)

↓

Lifestyle modifications, including smoking cessation and dietary advice—if needed from a certified dietitian

↓

Treatment priority:
- Children: statin, ezetimibe, and bile acid-binding resin
- Adults: maximal potent statin dose, ezetimibe, bile acid-binding resin, fibrate, (niacin, novel therapies)
- Lipoprotein apheresis in homozygotes and in treatment-resistant heterozygotes with coronary heart disease

→ **LDL cholesterol targets**:
- <3.5mmol/L (<135mg/dL) for children
- <2.5mmol/L (<100 mg/dL) for adults
- <1.8mmol/L (<70mg/dL) for adults with known coronary heart disease or diabetes

DUTCH LIPID CLINIC NETWORK DIAGNOSTIC CRITERIA FOR FAMILIAL HYPERCHOLESTEROLEMIA[1-3]

Criteria	Point
Family History	
First-degree relative with known premature* coronary and vascular disease OR First-degree relative with known LDL-C level above the 95th percentile.	1
First-degree relative with tendinous xanthomata and/or arcus cornealis OR Children aged less than 18 years with LDL-C level above the 95th percentile.	2
Clinical History	
Patient with premature* coronary artery disease.	2
Patient with premature* cerebral or peripheral vascular disease.	1
Physical Examination	
Tendinous xanthomata	6
Arcus cornealis prior to age 45 years	4
Cholesterol levels mg/dl (mmol/liter)	
LDL-C >= 330 mg/dL (≥8.5)	8
LDL-C 250 - 329 mg/dL (6.5-8.4)	5
LDL-C 190 - 249 mg/dL (5.0-6.4)	3
LDL-C 155 - 189 mg/dL (4.0-4.9)	1
DNA Analysis	
Functional mutation in the LDLR, apo B or PCSK9 gene	8
Diagnosis (diagnosis is based on the total number of points obtained)	
Defnite familial hypercholesterolemia	>8
Probable familial hypercholesterolemia	6 - 8
Possible familial hypercholesterolemia	3 - 5
Unlikely familial hypercholesterolemia	<3

* Premature = < 55 years in men; < 60 years in women
LDL-C = low density lipoprotein cholesterol; FH, familial hypercholesterolemia.
LDLR = low density lipoprotein receptor
Apo B = apolipoprotein B
PCSK9 = Proprotein convertase subtilisin/kexin type 9

[1]Austin MA, Hutter CM, Zimmern RL, Humphries SE. Genetic causes of monogenic heterozygous familial hypercholesterolemia: a HuGE prevalence review. American journal of epidemiology. 2004;160:407-420.

[2]Haase A, Goldberg AC. Identi cation of people with heterozygous familial hypercholesterolemia. Current opinion in lipidology. 2012;23:282-289.

[3]Nordestgaard BG, Chapman MJ, Humphries SE, et al. Familial hypercholesterolaemia is underdiagnosed and undertreated in the general population: guidance for clinicians to prevent coronary heart disease: consensus statement of the European Atherosclerosis Society. European heart journal. 2013;34:3478-3490a.

Børge G. Nordestgaard EHJ 2013

Heterozygous familial hypercholesterolemia in Hong Kong Chinese. Study of 252 cases

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[c] Macau Institute for Applied Research in Medicine and Health, Macau University of Science and technology, Taipa, Macau
[d] School of Biomedical Sciences, The Chinese University of Hong Kong, Shatin, Hong Kong
[e] Department of Ophthalmology and Visual Sciences, The Chinese University of Hong Kong, Shatin, Hong Kong

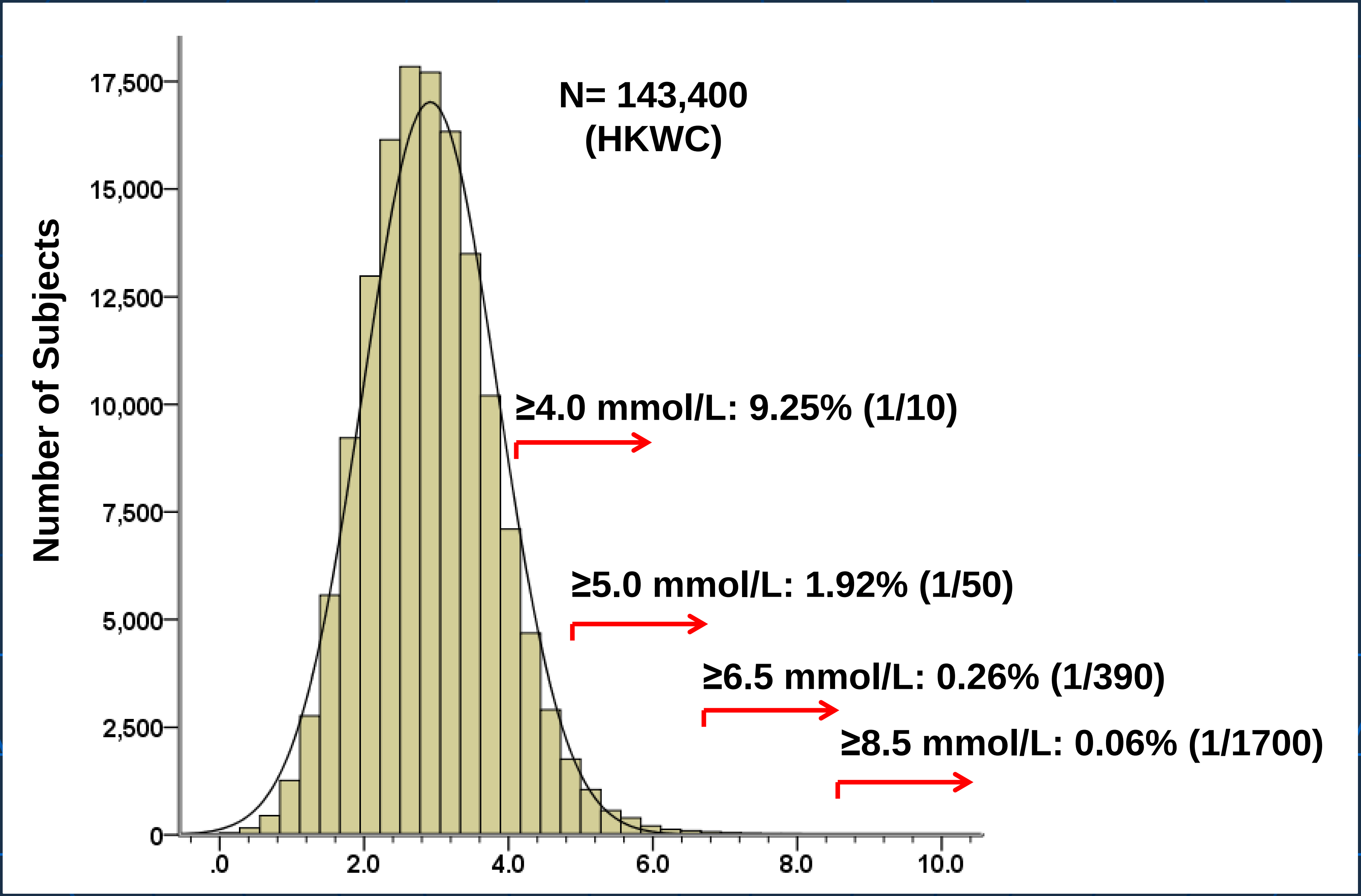
- From 1990 to 2000
- PWH (~1,000,000 population)
- TC > 7.5 mmol/L without secondary causes
- 87 patients diagnosed heterozygous FH

- Another 165 patients with heterozygous FH were identified through cascade screening

- 252 subjects were clinically diagnosed as heFH.

- Prevalence of heFH: 0.025%, i.e., 1/4000 populations

LDL-C in 143,400 Statin naïve Subjects in HK



SIU CW (unpublished data)

Questions?

Homework

1) Disease presentation, mechanisms and diagnostic criteria
    - Common connective tissue diseases
2) Pharmacology
    - Lipid lowering drugs: types, mechanisms of action, side effects